

Chapter 6

Computer Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

- To prevent potential data loss and equipment damage, please do the following:
- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 12](#) of this book.
 - Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

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Section 1.0

Introduction

This chapter describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

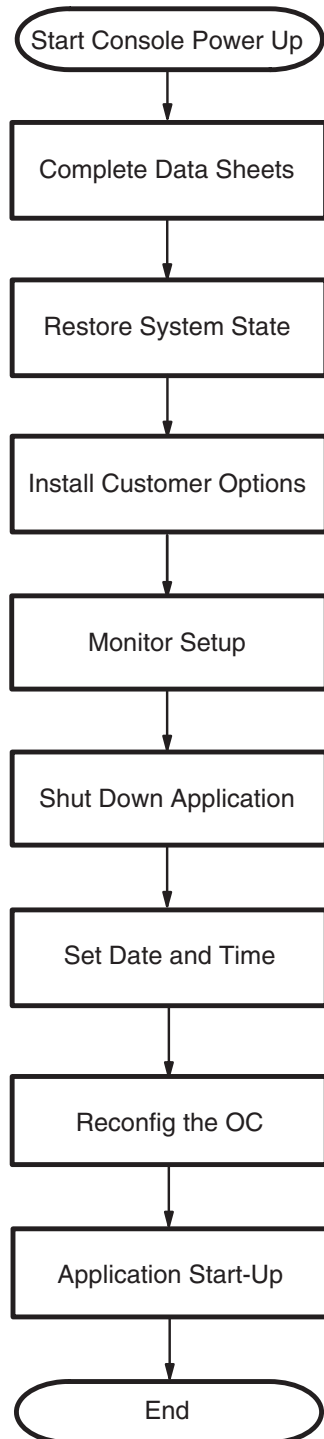


Figure 6-1 Computer Integration Process Overview

Section 2.0

System Configuration Data Sheets

2.1 Requirement

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their LightSpeed Plus scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 6-1 Manual Film Composer Options, on page 145](#)
- [Table 6-2 System Network Configuration, on page 146](#)
- [Table 6-3 Networking Application \(Image transfer\) Configuration, on page 147](#)
- [Table 6-4 DASM Laser Camera Configuration, on page 147](#)
- [Table 6-5 DICOM Print Camera Configuration, on page 148](#)
- [Table 6-6 DICOM Print Camera Advanced Configuration, on page 148](#)

2.2 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 6-1 Manual Film Composer Options

2.3 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 6-2 System Network Configuration

2.4 Host Ethernet Address

_____ : _____ : _____ : _____ : _____

2.5 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 6-3 Networking Application (Image transfer) Configuration

2.6 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 6-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION

Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 6-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION

Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 6-6 DICOM Print Camera Advanced Configuration

Section 3.0 Configuration

Note: Type the text shown in **boldface**, and press the ENTER key on the keyboard. (Type/enter bold = Type bold ENTER.)

3.1 Restore System State

Tools Required: None

Your system should have a system state MOD, located in the software collector box.
The system state MOD contains:

- Collimator Characterization
- Phantom Calibrations

The installation process uses all the system state files. At this time, use the system state MOD to restore the phantom calibrations and system characterization files.

If you cannot locate an existing system state MOD, manually characterize the system and perform phantom calibrations.

1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the PM icon.

3.) Select SYSTEM STATE.

4.) Insert the MOD in the MOD drive.

5.) Select CHARACT.

6.) Select CALS.

7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Note: **Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error should occur during the restore process, see the Software Load Procedure manual (2211220-100) for information regarding possible error messages and their recovery.

8.) Click NO for Reset Scan Hardware popup message.

9.) Select DISMISS.

3.2 Install Customer Options

3.2.1 Software

TOOLS REQUIRED

None

PROCEDURE

Note: **Your system may have a MOD that contains customer purchased options. If your system has an options MOD, install it at this time.**

Ensure that the options MOD is NOT write protected at this time. The initial install requires that the MOD be write enabled; subsequent installs can be done with the MOD write protected.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select INSTALL OPTIONS.

An Options Window appears (Figure 6-2):

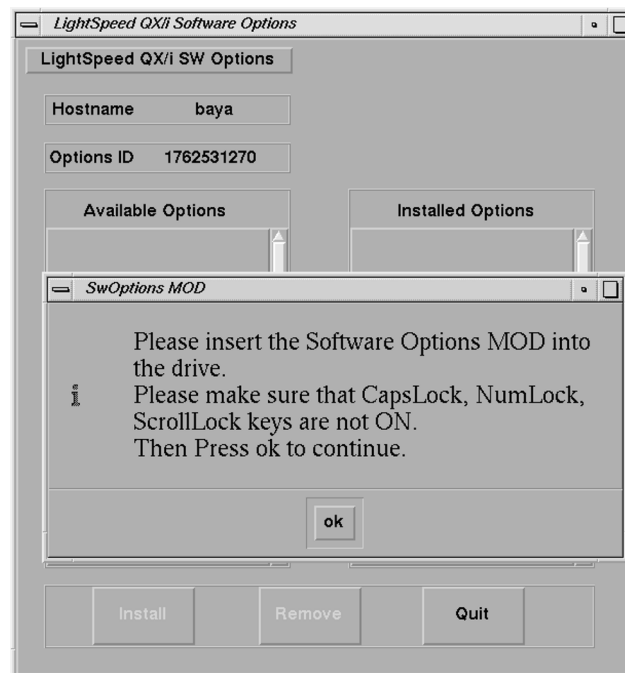


Figure 6-2 Options Window when First Selected

- 5.) Insert the options MOD into the MOD drive and click on OK. (If you do not have an options MOD, click on OK anyway, wait for the abort pop up, then abort the process.)

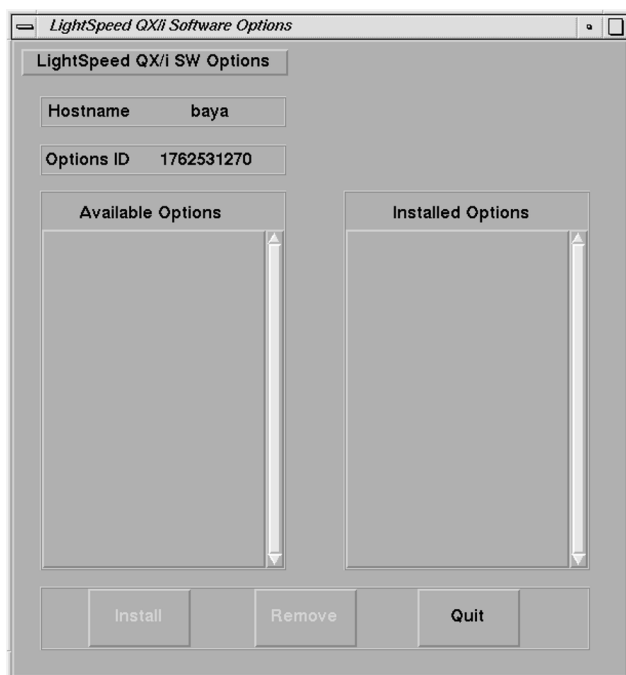


Figure 6-3 Example: Options Window

- 6.) Select all of the options in the left-hand column to install the corresponding software.
- 7.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the Installed Options list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).
- 8.) After the options are installed, select QUIT.
- 9.) Select OK.
- 10.) Remove the MOD and write protect the side with options.
- 11.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

3.2.2 Camera

TOOLS REQUIRED

Small flat-blade screwdriver

PROCEDURE

Note: DASM hardware must be installed before proceeding.
For details on camera configuration, refer to the Software Load Procedures manual.
For details on troubleshooting the camera, refer to the System Service manual.

- 1.) Click on the SERVICE DESKTOP icon.
- 2.) Select UTILITIES.
- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to Home Page.

3.3 Monitor Setup

3.3.1 Overview

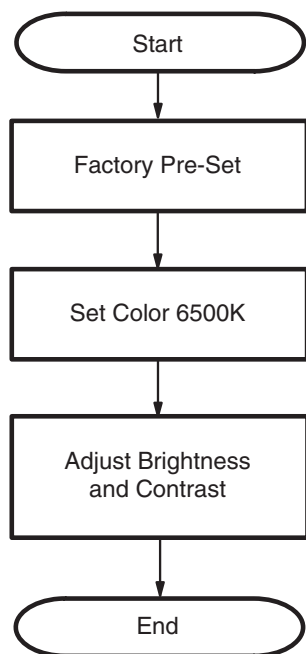


Figure 6-4 Monitor Setup Flowchart

3.3.2 Tools Required

Sony Service Manual (located in monitor box)

3.3.3 Color Adjustments

- 1.) Adjust the ambient room light to the level used by the operator during image review.
- 2.) Select the COLOR menu to adjust the picture's color temperature. (You can use this to match the monitor's colors to a printed picture's colors.

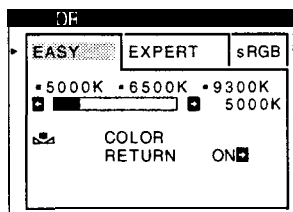


Figure 6-5 Color Menu

- 3.) Set monitor color temperature to 6500K. Refer to the system service manual for this procedure, if necessary.
- 4.) Use one of the following methods:

- a.) Click the IMAGEWORKS icon  and proceed to step 3.
- OR**

b.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.

Click on the IMAGE QUALITY icon.



Select INSTALL SMPTE IMAGE and proceed to step 3. (Approximately 3-4 minutes.)

5.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.

6.) Increase the Contrast to maximum.

Note: Adjust monitor contrast until the operator sees the anatomical structure (window width) at the expected window levels.

7.) Increase the Brightness to maximum.

Note: Adjust the monitor brightness until the operator sees the anatomical structure (window width) at the expected window levels.

8.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.

If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.

If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.

9.) To view SMPTE on text monitor, swap video cable.

3.3.4 Using Menu Buttons

1.) Press the Main Menu button to display the main menu on your screen.



Figure 6-6 Main Menu Button

2.) Press the / buttons to highlight the menu you want to adjust. Press the Main Menu button to select the menu item.

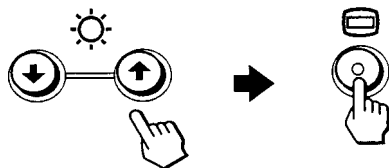


Figure 6-7 Select Menu Item

- 3.) Press the  buttons to select the desired adjustment item. Press the  buttons to make the adjustment.

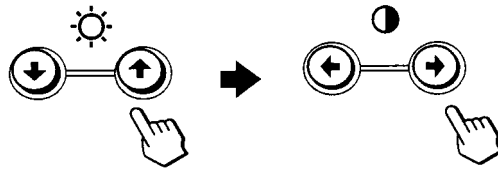


Figure 6-8 Adjust the Menu

- 4.) Press the Main Menu button once to return to the main menu, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.



Figure 6-9 Main Menu Button

- 5.) Press the reset button to reset the adjustments back to the factory settings.



Figure 6-10 Reset Button

3.3.5 Adjusting Brightness and Contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu. These settings are stored in memory for the signals from the currently selected input connector.

- 1.) Press either one of the brightness or contrast buttons. The BRIGHTNESS/CONTRAST menu appears on the screen.

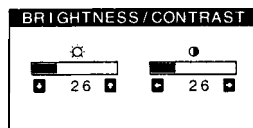


Figure 6-11 Brightness/Contrast Menu

- 2.) Press the  buttons to adjust the brightness and the  buttons to adjust the contrast.

3.4 Shut Down Application

TOOLS REQUIRED

None

PROCEDURE

- 1.) If Applications is currently running, you must shutdown system applications to the CT user desktop.
 - a.) Click on the SERVICE DESKTOP button.
 - b.) On the desktop toolbar select UTILITIES.
 - c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
- 2.) Open a UNIX Shell window.
 - a.) Type: `su` - ENTER at the prompt.
 - b.) Enter the root (super user) password: `#bigguy`

3.5 Check/Set Date and Time

TOOLS REQUIRED

None

PROCEDURE

Check the date and time.

- 1.) Type/enter `setdate` ENTER.
- 2.) Follow the instructions to set the correct time and date.

3.6 Reconfig the OC

3.6.1 Tools Required

None

Note: The document collector box that arrived with your system contains the LightSpeed Plus Software Load Procedure manual, which documents the reconfiguration procedure in more detail.

3.6.2 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWHHMMSS
```

Where

YYYYMMDDWWHHMMSS is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

3.6.3 Procedure

- 1.) Change directory to scripts:
 - a.) Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the LightSpeed Install utility:
 - a.) Type: `reconfig` ENTER at the prompt.

The OC displays the LightSpeed Install Utility Window as shown in [Figure 6-12](#).

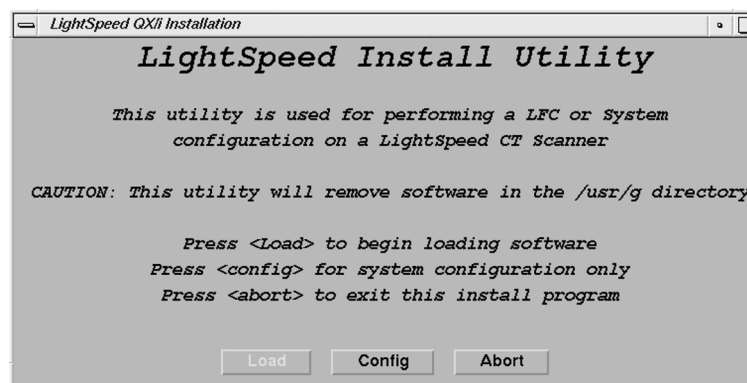


Figure 6-12 Install Utility Window

- 3.) Using the mouse, click on the CONFIG button.

The OC displays the LightSpeed Plus System Configuration - System Settings Screen as shown in [Figure 6-13](#).

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

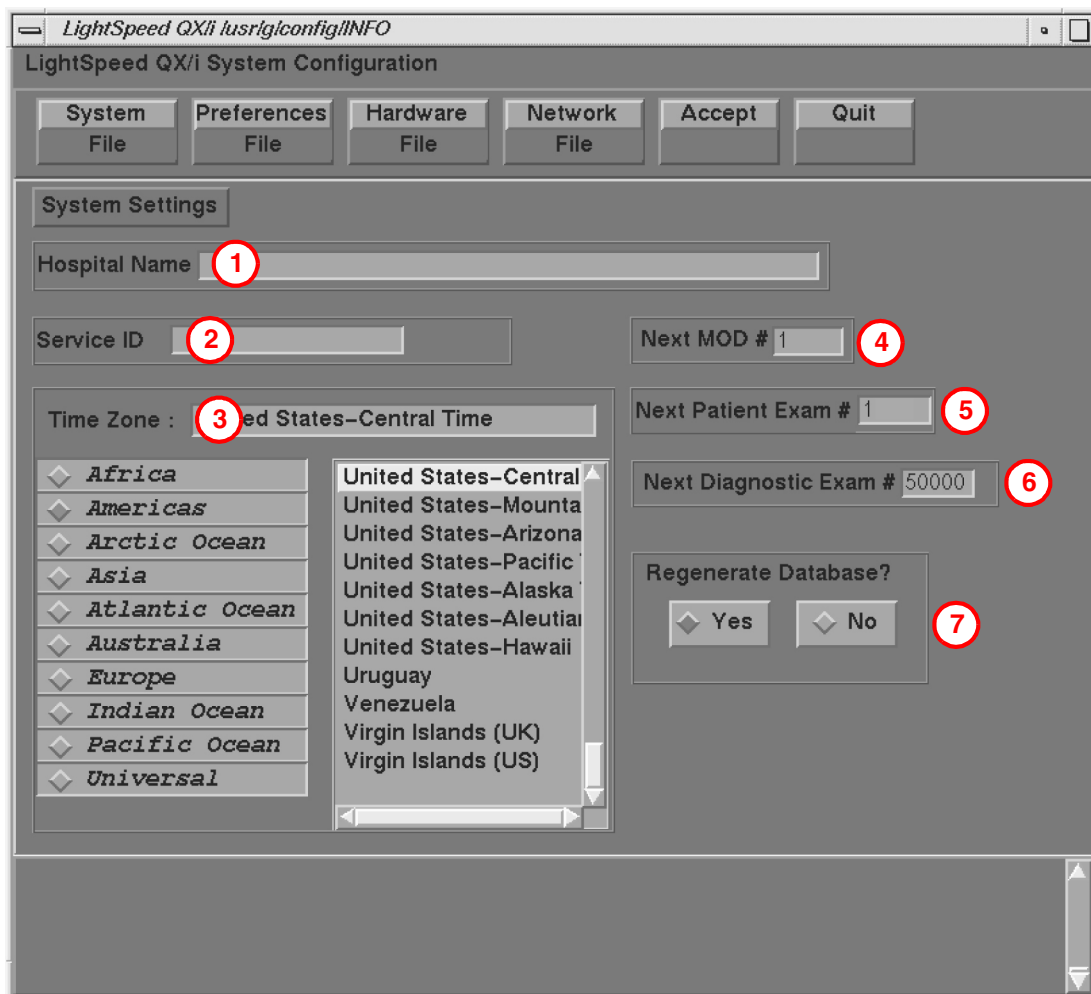


Figure 6-13 System Settings Screen

4.) Configure System Settings

- a.) Hospital Name (Figure 6-13, item 1) configures the name that will show up on images produced by this scanner. *Example: ST MARYS HOSPITAL*
- b.) Service ID (Figure 6-13, item 2) is issued by the Service organization. *Example: 262785CT2 (no spaces)*
- c.) Select the Time Zone for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
- e.) Next Patient Exam # configures the next Exam number the scan user interface will

- use. At initial system installation, type: 1
- f.) Next Diagnostic Exam # *Customer Selected*.
- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in [Figure 6-14](#).
 - 6.) Regenerate Database (refer to [Figure 6-13](#)) determines whether the Scan Database will be regenerated during reconfiguration.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

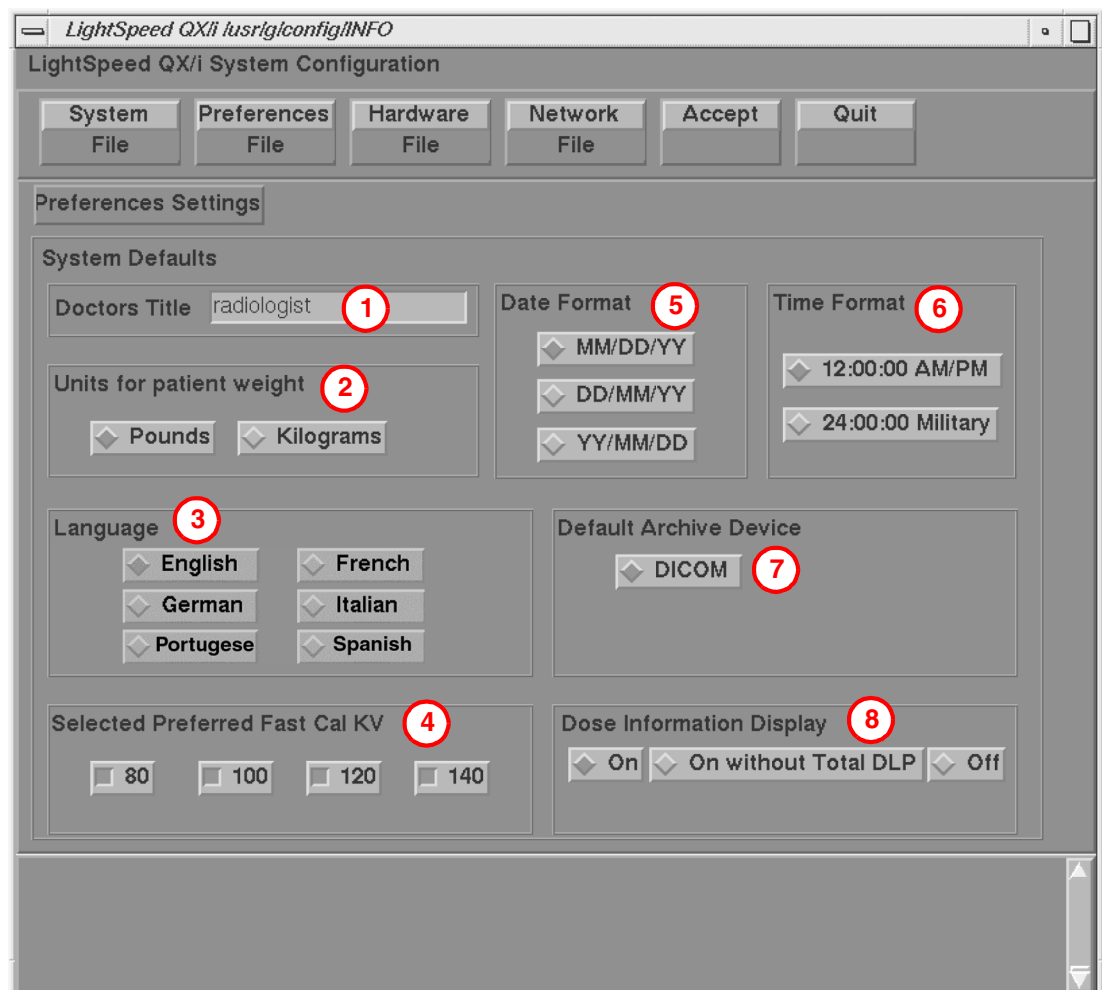


Figure 6-14 Preferences Setup Screen

7.) Configure Preferences Settings

- a.) Language configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGUESE, SPANISH).
- b.) Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.
Comment: These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs
- c.) Date Format and Time Format configure the format in which the date and time will be displayed on the images.
- d.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)

- 8.) Select the HARDWARE button to display the Hardware Settings Screen (Figure 6-15).

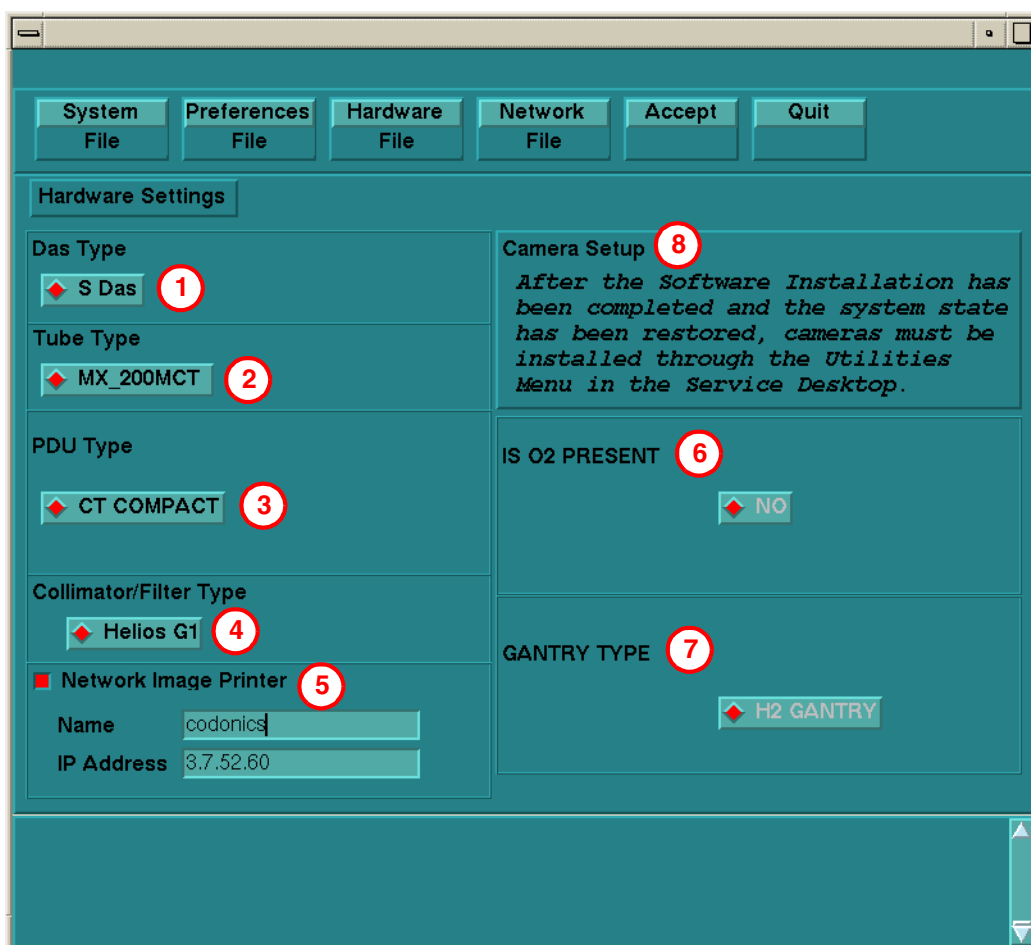


Figure 6-15 Hardware Settings Screen

9.) Configure Hardware Settings

- a.) DAS Type configures which DAS the software should expect.
- b.) Tube Type configures which X-Ray Tube the software should expect.
- c.) PDU Type configures which PDU the software should expect.
- d.) Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, LightSpeed Plus systems only support HELIOS G1.
- e.) Gantry Type selects the type of Gantry connected to this console.
- f.) Select Network Image Printer only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system. When selected:
 - * Enter the printer Hostname in the Name field.
 - * Enter the printer IP Address in the IP Address field.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

g.) Camera Setup:

Comment: Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started. See "Declaring DICOM Filming Devices on the LightSpeed System" on page 255.

10.) Select the NETWORK button to display the Network Settings Screen as shown in [Figure 6-16](#).

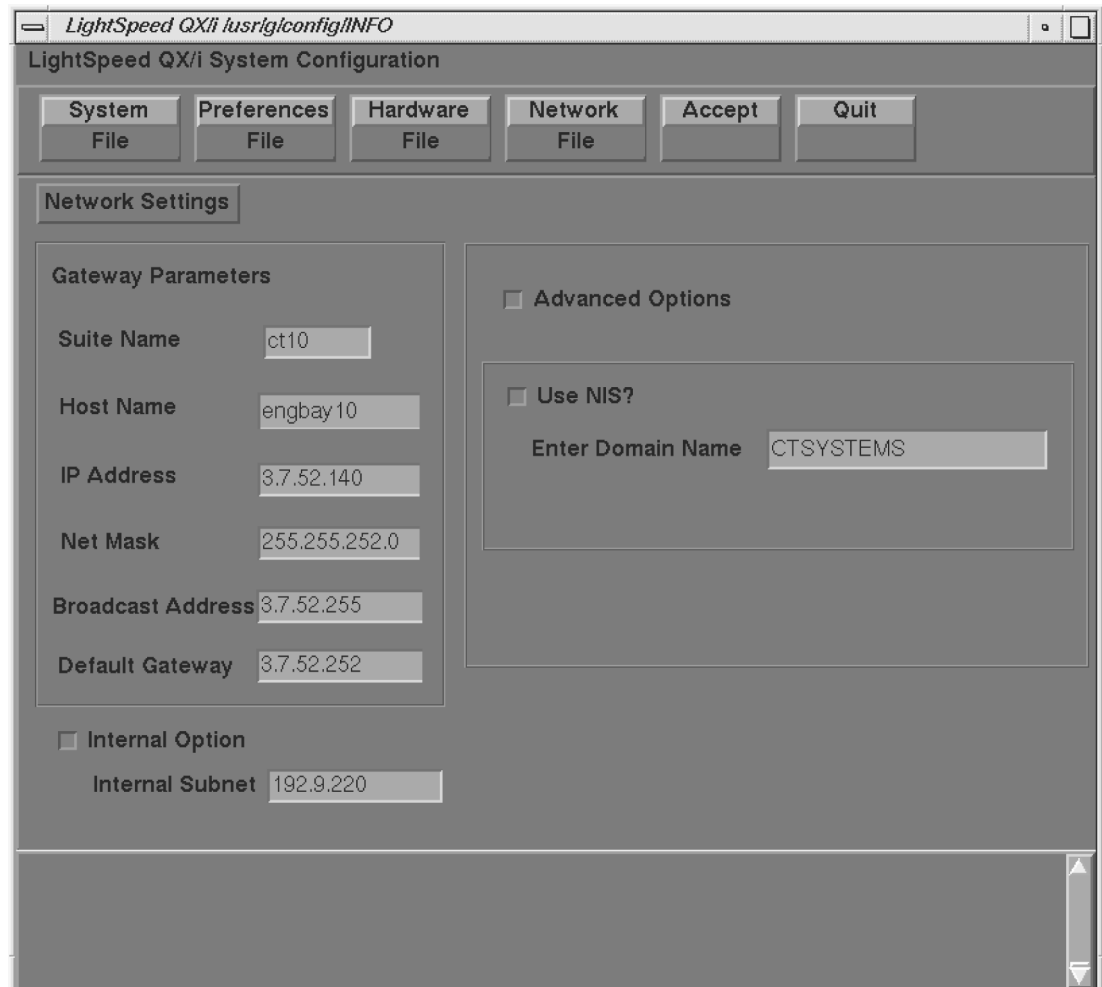


Figure 6-16 Network Settings Screen

11.) Configure Network Settings

Comment: This screen provides the ability to declare the LightSpeed system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for LightSpeed systems on a subnet) must be obtained from the hospital network administrator.

See Chapter 13 for more information and complete details of setting the Hospital/System Network Configuration.

a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the LightSpeed system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment: The Host Name is typically stmary or ct01.

c.) Enter the hospital provided IP Address for the system.

d.) Enter the hospital provided Net Mask (if the LightSpeed system is on a subnet).

e.) Enter the Broadcast Address.

Comment: The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.

g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:

- * Select ADVANCED OPTIONS button on the Network Settings Screen.
- * Select USE NIS button.
- * Enter the hospital provided Domain Name (from the hospital network administrator).

12.) Select ACCEPT on the LightSpeed Plus System Configuration Screen.

Comment: The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named:

- 13.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 6-17](#).)



Figure 6-17 Reboot Screen

- 14.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

3.7 Applications Start-Up

Open the Console shell window, and type: `st` ENTER. The applications desktop appears on the monitor.